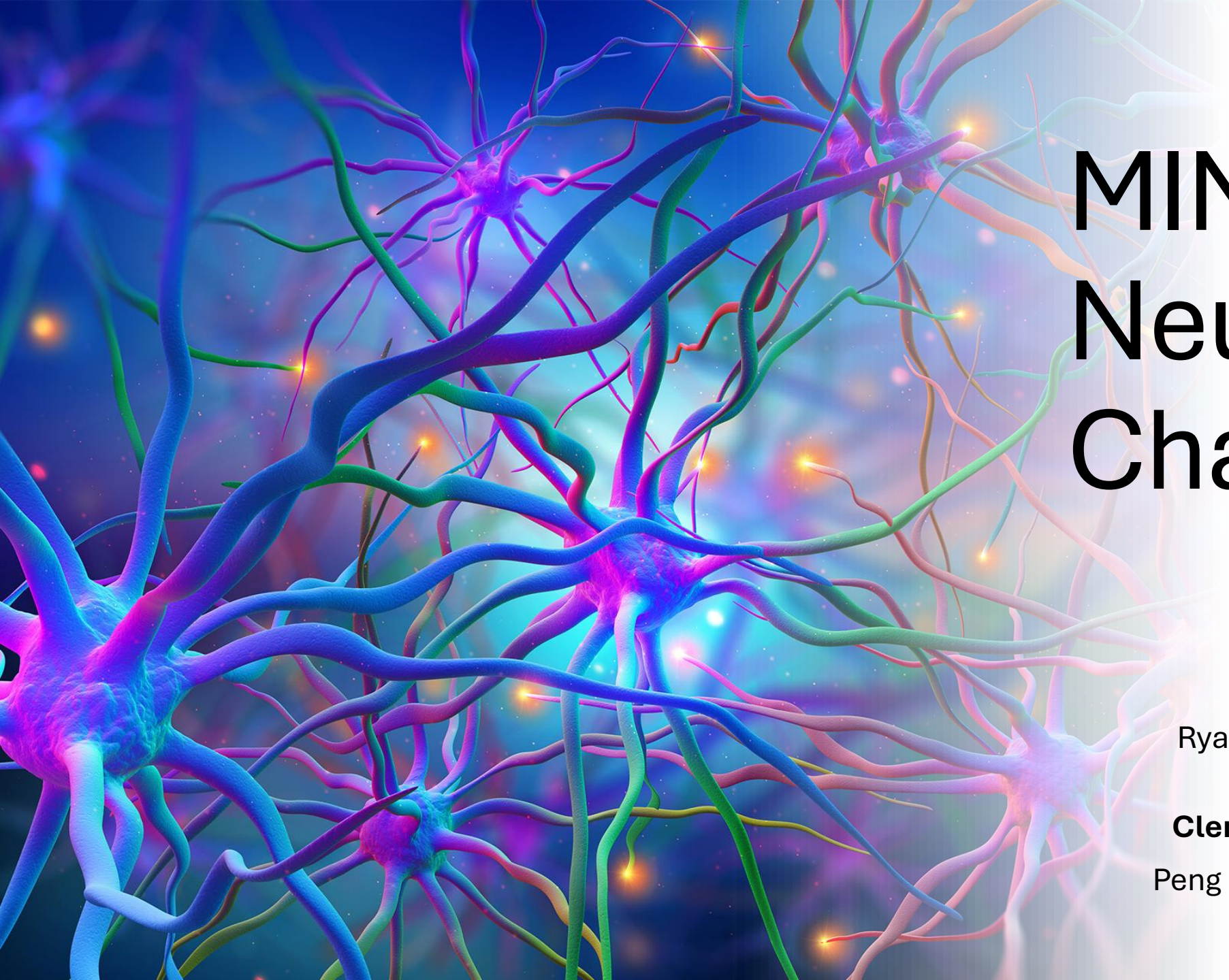

MIND Neurocoding Challenge

Team Meowbunga

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Clemson University Bioengineering

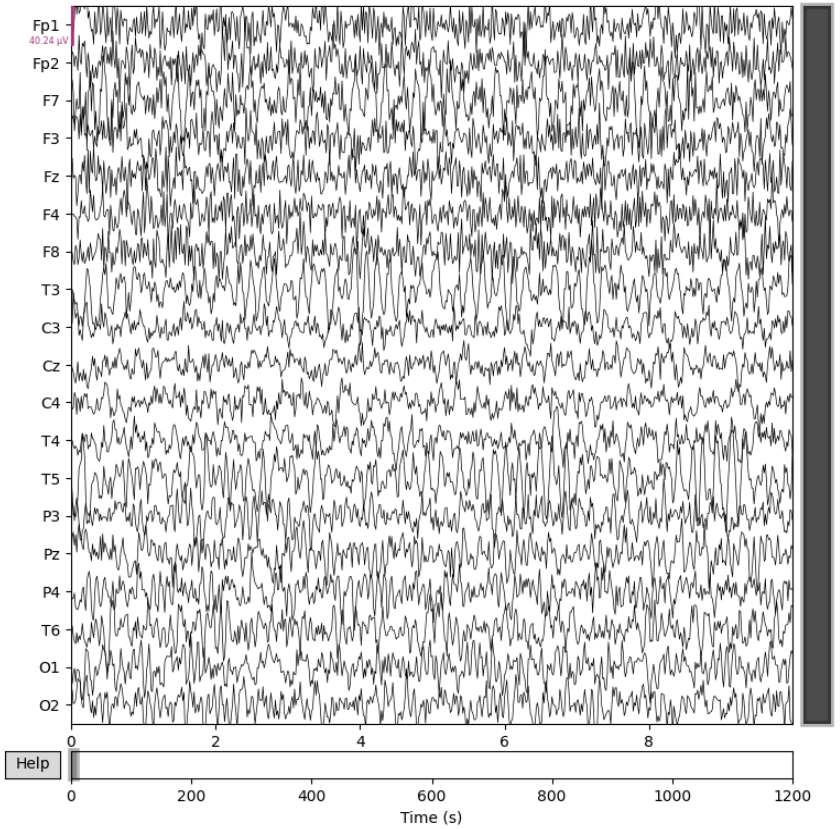
Peng Lab, advised by Dr. Congyue Annie Peng

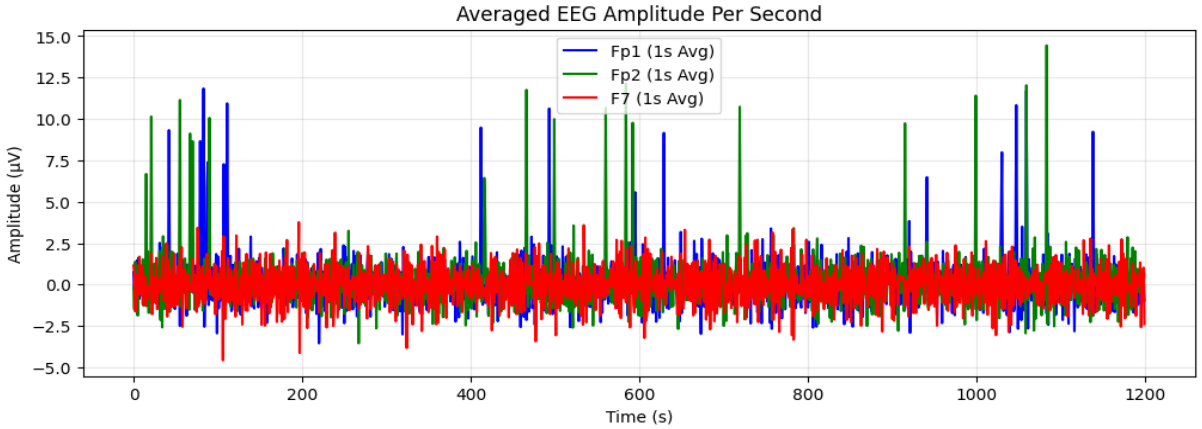
Study Case

- Seizures began at age 19
- Episodes of “becoming absent” (loss of awareness)
- Tried multiple antiepileptic medications
- Underwent surgical resection → temporary improvement
- Seizures returned

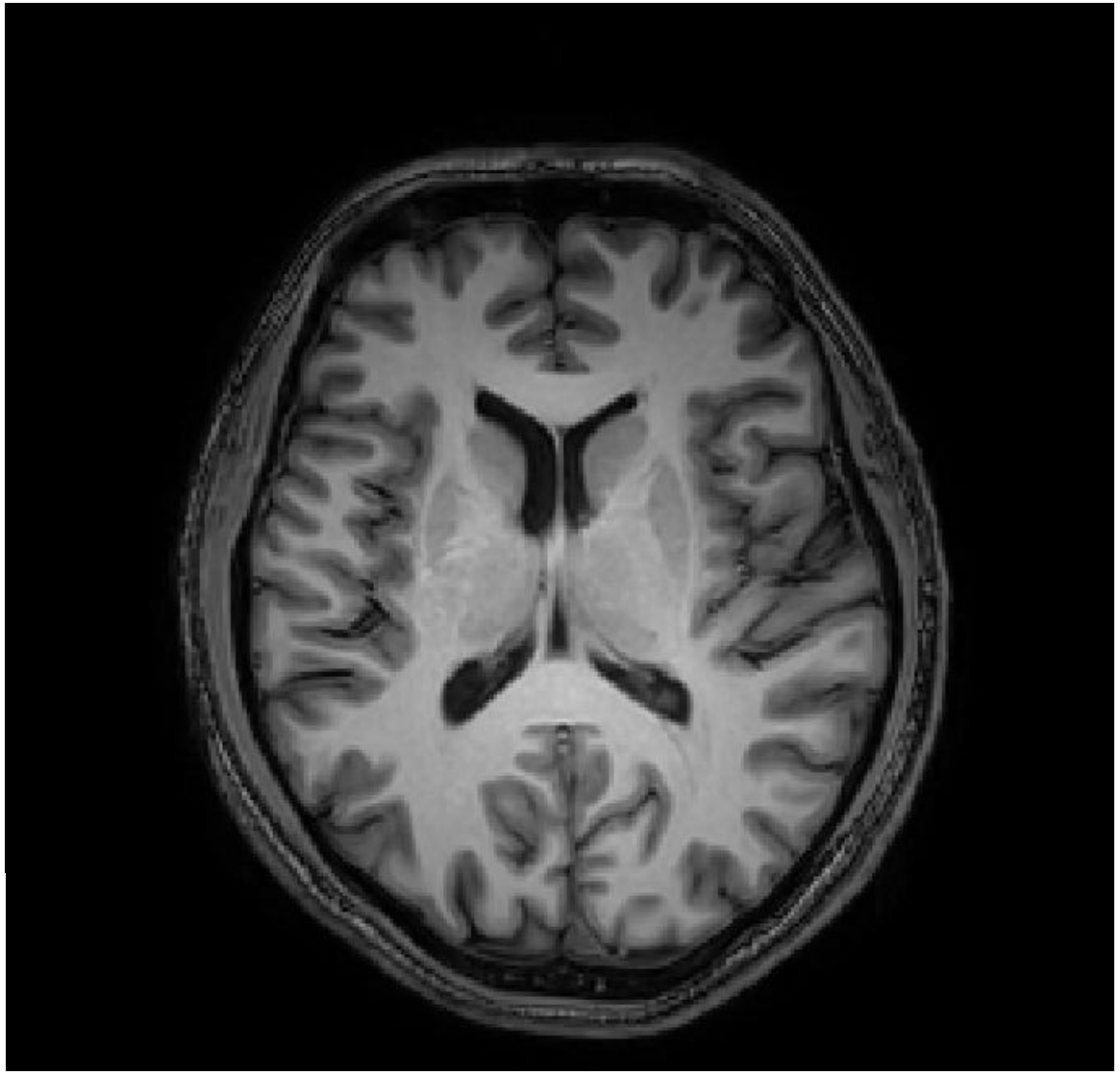
20 Minute Scalp EEG Analysis

- EEG data stored in CSV file.
- Use of MNE (Minimum Norm Estimate) Python Package a standard open-source package for reading and analyzing EEG Data.
- Platform: Google Collaboratory, Visual Studio Code (VSCode).
- Result: No obvious epileptiform activity.





Brain MRI

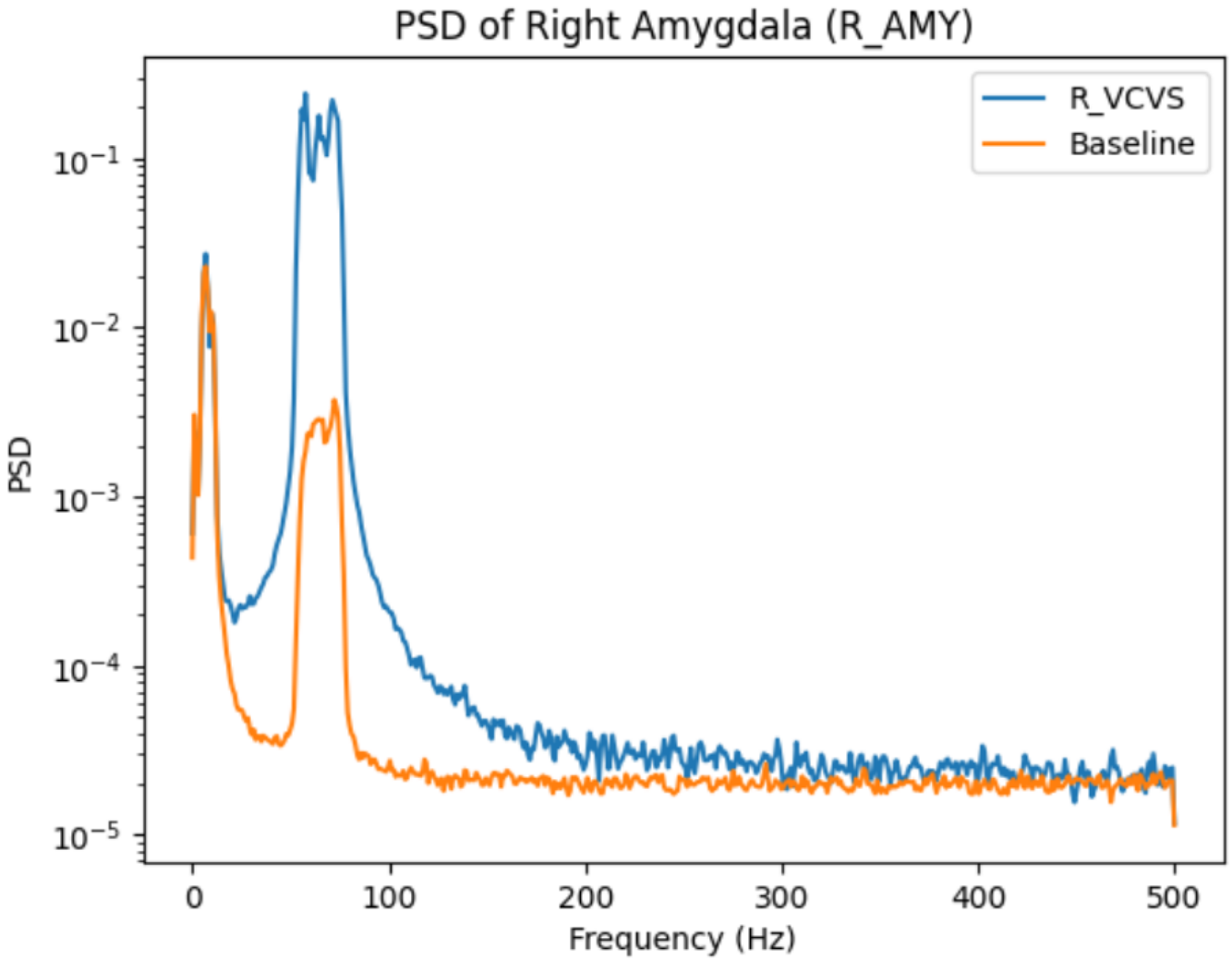


```
>> % NIFTI UI Slider
filepath = 'sub-001_ses-2024110601_acq-2mm_T1w.nii';
V = double(niftiread(filepath));
vol = rot90(V,1);
numSlices = size(vol,3);

fig = uifigure('Name','NIfTI Viewer');
ax  = uiaxes(fig, 'Position',[20 0 560 560]); colormap(ax,gray)
h = imshow(vol(:,:,1), [], 'Parent', ax);

s = uislider(fig, 'Limits',[1 numSlices], 'Value',1, 'Position',[80 30 420 3]);
s.ValueChangingFcn = @(src,ev) set(h,'CData',vol(:,:,round(ev.Value)));
>>
```

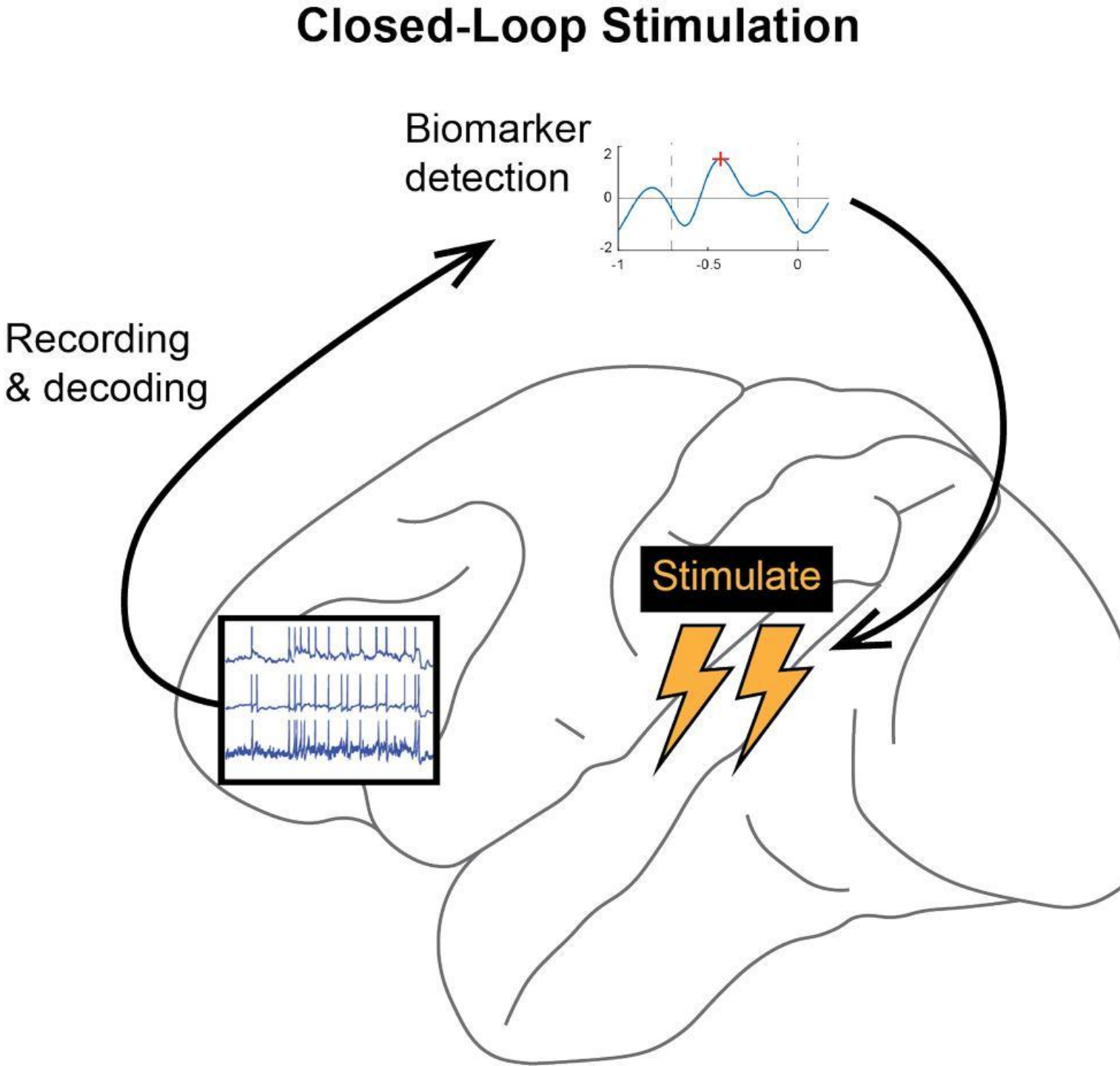
sEEG analysis on the VC/VS Region



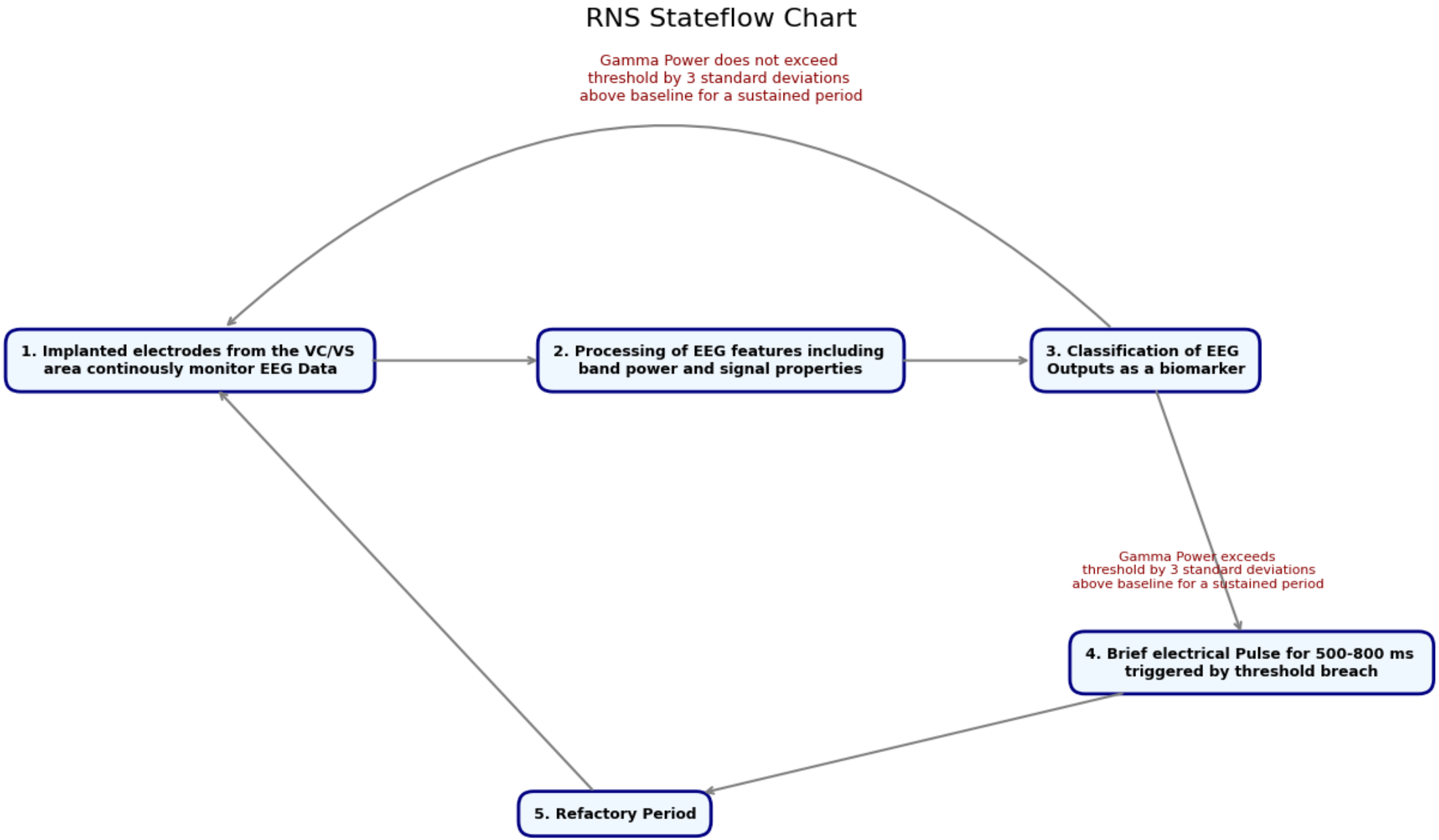
Solution: Closed-Loop Neuromodulation

Concept:

- Monitor brain signals in real time
- Detect seizure-related or abnormal states
- Deliver stimulation **only when needed**
- **System components:**
- Inputs: neural biomarkers (EEG signals)
- Decision: classify brain state
- Output: adjust stimulation



Stateflow Chart of Responsive Neural Stimulation



Thank you!